Power In Waves

> SYNOPSIS
>
> We round up the notes with equations for calculating power in waves.

BEFORE YOU READ ON — PREDICT

You are standing in an open field listening to a loudspeaker, and you walk until you are **twice as far away**. **(1)** What happens to the *intensity*? **(2)** What happens to the *amplitude* of the wave? **(3)** By how many *decibels* does the level drop? Write down your three answers. This note gives you all three, and the third one is a number worth carrying around.

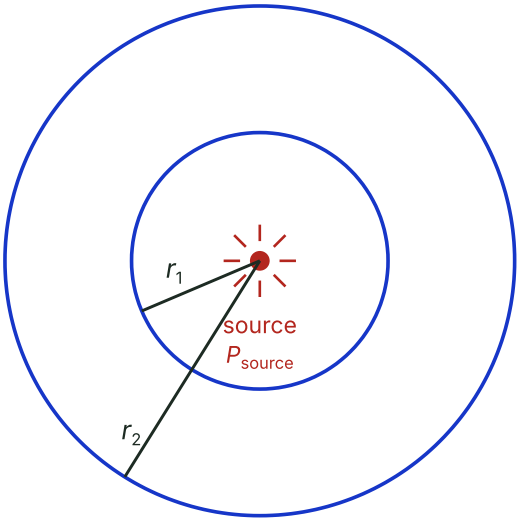
PERFORMANCE EXPECTATIONS

1. **Define** power and intensity, and give the SI unit of each.
2. **Derive** $I = P/4\pi r^2$ from the area of a sphere, and say what is *not* being used up as the wave spreads.
3. **Explain** why intensity goes as the square of the amplitude.
4. **Define** the decibel against its reference I_0, and convert an intensity ratio into a level difference.
5. **State** what a 10 dB rise means physically, and what level a sound at the reference intensity has.

3. Power in Waves

Note 10 counted the energy an oscillator carries. This note follows that energy outwards: how fast it leaves the source, how thinly it is spread by the time it reaches you, and why we end up measuring it on a logarithmic scale.

1. Write **power** and **intensity** as definitions, and give the SI unit of each.
2. A source radiates P_{source} equally in every direction. At a distance r, what area is that power spread over? Write $I(r)$.
3. Write I_1 and I_2 at two distances and form the ratio I_1/I_2. What cancels, and what does that tell you?
4. Intensity turns out to go as the **square of the amplitude**. Given $E = \tfrac{1}{2}kA^2$ for a sinusoidal wave, why is that the relationship you should have expected?



Label the intensity on each sphere and write it in terms of P_{source} and the radius. What is the area of a sphere, and what cancels when you divide one intensity by the other?

THE PUNCHLINE

The **inverse-square law**, and notice what it is *not*: nothing here is being used up. The source puts out the same power whether you stand near it or far from it; all that happens is that the power gets spread over a bigger and bigger sphere. Every inverse-square law in physics — sound, light, gravity, the electric field — is this same piece of geometry, which is why the exponent is exactly 2 and not 1.9.

Sound intensity level (decibel)

A quiet room and a jet engine differ in intensity by about a factor of a trillion. No linear scale is going to show both ends of that comfortably, so sound is measured logarithmically, against a fixed reference.

5. Write the definition of the sound intensity level in decibels, and say what the reference I_0 is.
6. What does an increase of **10 dB** mean, in terms of intensity? And what level does a sound at the reference intensity itself have?

Looking ahead

Everything so far has had the source and the listener standing still. **Note 12** sets them moving: when a source or an observer is travelling, the frequency that arrives is not the frequency that left.

Where this sits. Note 10 found the energy stored in one oscillator and it went as A^2. This note sends that energy out into space, and the A^2 survives the trip: intensity goes as amplitude squared at every distance. The $1/r^2$ is not physics on top of that — it is the geometry of a sphere, and it would be true of anything spreading evenly in three dimensions.

BACK TO YOUR PREDICTION

(1) The intensity drops to a quarter. $I \propto 1/r^2$, so doubling r divides I by 4 — while the *power* of the loudspeaker has not changed at all. **(2) The amplitude halves.** $I \propto A^2$, so a factor of 4 in intensity is a factor of 2 in amplitude. **(3) Down 6 dB** — $(10\text{ dB})\log_{10}(¼) = -6.02$ dB. **Every doubling of distance costs you 6 dB**, whatever the source and whatever the starting level, which is the one number from this note worth remembering by itself.